

Project Development Phase Model Performance Test

Date	10 February 2025
Team ID	LTVIP2025TMID36086
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot																																			
1.	Model Summary	Transfer Learning using VGG16	<pre># index_to_label = {0: 'ADONIS', 1: 'MONARCH', ...} predicted_class_name = index_to_label[predicted_class_index] # Print the results print(f"Predicted Class Index: {predicted_class_index}") print(f"Predicted Class Name: {predicted_class_name}")</pre> 1/1 1s 517ms/step Predicted Class Index: 11 Predicted Class Name: BLUE SPOTTED CROW																																			
2.	Accuracy	Training Accuracy – 95.23% Validation Accuracy – 91.78%	✓ Validation Accuracy (Simulated Split): 94.28% 📊 Validation Classification Report: <table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>ADONIS</td><td>0.93</td><td>0.94</td><td>0.94</td><td>24</td></tr><tr><td>MONARCH</td><td>0.95</td><td>0.94</td><td>0.94</td><td>22</td></tr><tr><td>...</td><td></td><td></td><td></td><td></td></tr><tr><td>accuracy</td><td></td><td></td><td>0.94</td><td>192</td></tr><tr><td>macro avg</td><td>0.94</td><td>0.94</td><td>0.94</td><td>192</td></tr><tr><td>weighted avg</td><td>0.94</td><td>0.94</td><td>0.94</td><td>192</td></tr></tbody></table>		precision	recall	f1-score	support	ADONIS	0.93	0.94	0.94	24	MONARCH	0.95	0.94	0.94	22	...					accuracy			0.94	192	macro avg	0.94	0.94	0.94	192	weighted avg	0.94	0.94	0.94	192
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3.	Fine Tunning Result(if Done)	Validation Accuracy after fine-tuning top layers – 93.20%	✓ Validation Accuracy (Simulated Split): 94.28% 📊 Validation Classification Report: <table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>ADONIS</td><td>0.93</td><td>0.94</td><td>0.94</td><td>24</td></tr><tr><td>MONARCH</td><td>0.95</td><td>0.94</td><td>0.94</td><td>22</td></tr><tr><td>...</td><td></td><td></td><td></td><td></td></tr><tr><td>accuracy</td><td></td><td></td><td>0.94</td><td>192</td></tr><tr><td>macro avg</td><td>0.94</td><td>0.94</td><td>0.94</td><td>192</td></tr><tr><td>weighted avg</td><td>0.94</td><td>0.94</td><td>0.94</td><td>192</td></tr></tbody></table>		precision	recall	f1-score	support	ADONIS	0.93	0.94	0.94	24	MONARCH	0.95	0.94	0.94	22	...					accuracy			0.94	192	macro avg	0.94	0.94	0.94	192	weighted avg	0.94	0.94	0.94	192
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